ESTRUTURAS DE CONCRETO ARMADO II

Aula 01b — 20 de agosto de 2026

Exercício

1. Considerando a planta de forma estrutural apresentada, pede-se:

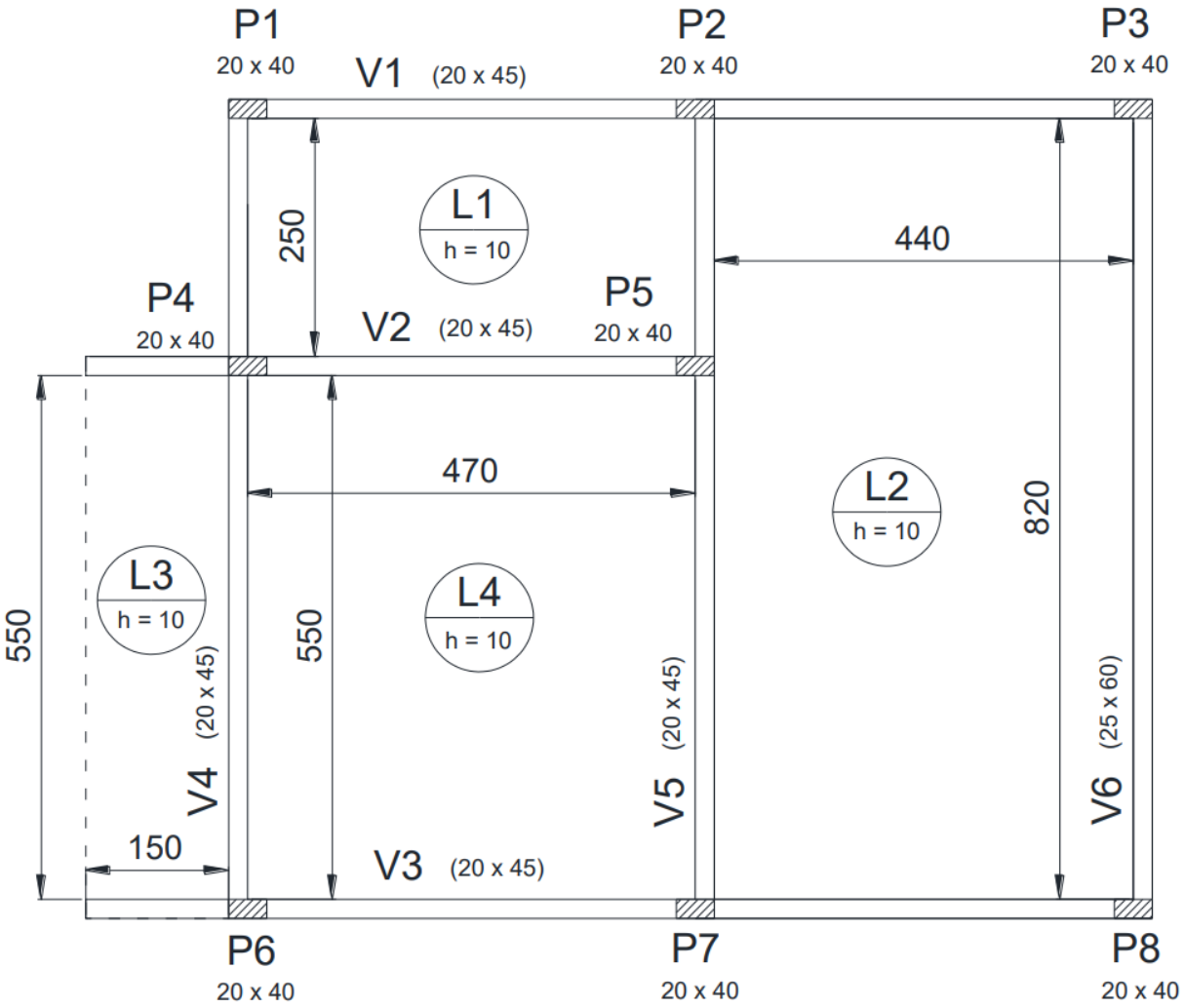


Figura 1 – Forma pavimento tipo.

- classificar as lajes em armadura me uma ou duas direções;
- indicar a vinculação de cada uma das lajes;
- considerando carregamento permanente $g = 3,5\ kN/m^2$ (peso próprio da laje + peso próprio do piso e revestimentos) e carregamento variável normal $q = 3\ kN/m^2$ (edifícios comerciais), calcular as reações de apoio das lajes.

- considerando carregamento permanente $g = 3,5\ kN/m^2$ e carregamento variável normal $q = 3\ kN/m^2$, calcular e compatibilizar os momentos atuando nas lajes.

1. Determinação dos vãos efetivos e classificação

l_x menor dimensão

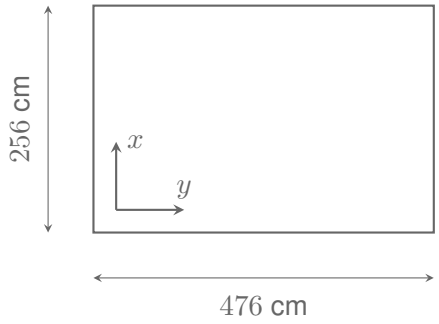
l_x maior dimensão

b_{w_0} zero é a indicação da viga

$\lambda \leq 2$ armada em duas direções

$\lambda > 2$ armada em uma direção

L1



a) Direção horizontal

$$l_0 = 470\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_4}}{2} = \dfrac{20}{2} = 10\ cm \\ 0,3h = 0,3 \times 10 = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 476\ cm$$

b) Direção vertical

$$l_0 = 000\ cm$$

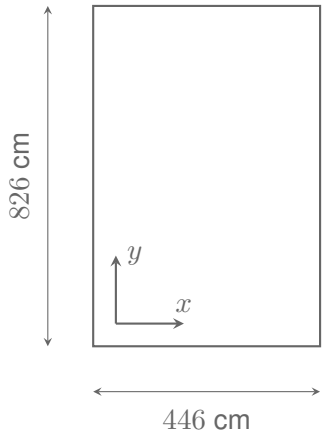
$$a_1 \leq \begin{cases} \dfrac{b_{w_1}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 256\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{476}{256} \approx 1,86 \leq 2 \rightarrow \text{armada em duas direções}$$

L2



a) Direção horizontal

$l_0 = 440\ cm$

$$a_1 \leq \begin{cases} \frac{b_{w_1}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \frac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_{ef} = l_0 + a_1 + a_2 = 446\ cm$

b) Direção vertical

$l_0 = 820\ cm$

$$a_1 \leq \begin{cases} \frac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

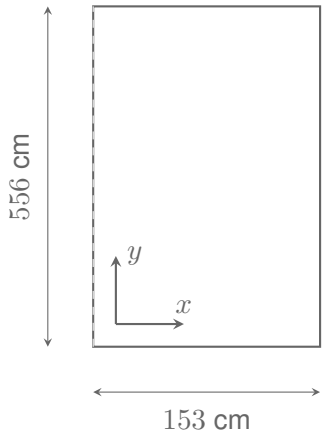
$$a_2 \leq \begin{cases} \frac{b_{w_6}}{2} = \frac{25}{2} = 12,5\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_{ef} = l_0 + a_1 + a_2 = 826\ cm$

$\lambda = \frac{l_y}{l_x} = \frac{826}{446} \approx 1,85 \leq 2 \rightarrow$ armada em duas direções

Continuação - para casa

L3



a) Direção horizontal

$$a_1 \leq \begin{cases} \frac{b_{w_4}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_0 = 150\ cm$

$$l_{ef} = l_0 + a_1 + a_2 = 153\ cm$$

b) Direção vertical

$$l_0 = 550\ cm$$

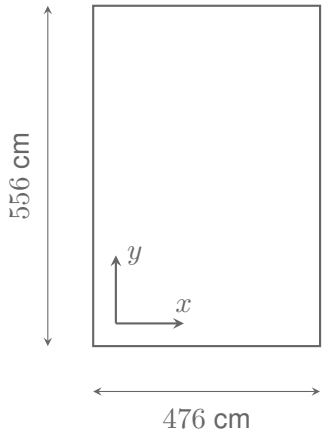
$$a_1 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 556\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{556}{153} \approx 3,63 > 2 \rightarrow \text{armada em uma direção}$$

L4



a) Direção horizontal

$$l_0 = 470\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 476\ cm$$

b) Direção vertical

$$l_0 = 550\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_4}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

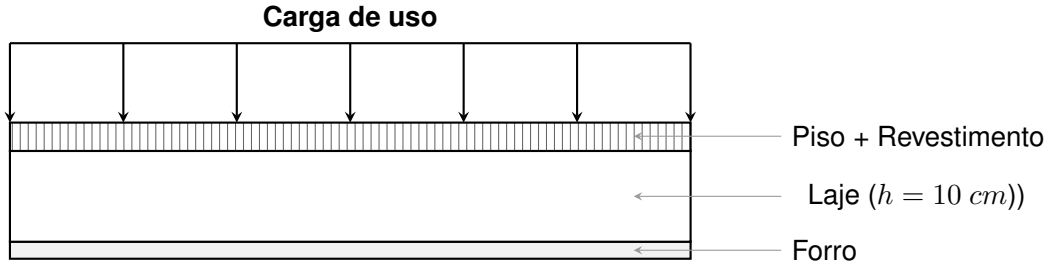
$$l_{ef} = l_0 + a_1 + a_2 = 556\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{556}{476} \approx 1,17 \leq 2 \rightarrow \text{armada em duas direções}$$

Aula 02 — 26 de agosto de 2026

2. Determinação do carregamento nas lajes

   É preciso calcular o carregamento permanente para cada laje, mas como h = 10 cm para todos não precisou repetir.



$g_{PP} = \gamma_{ap} \cdot h = 25 \times 10 \cdot 10^{-2} = 2,5\ kN/m^2 \rightarrow \gamma_{ap} = 25\ kN/m^3$ (**??**, tabela 1)

$g_{P+R} = 1\ kN/m^2$ (**??**, tabela 4)

$g_{forro} = 0,25\ kN/m^2$ (**??**, tabela 8)

$g = g_{PP} + g_{P+R} + g_{forro} = 2,5 + 1 + 0,25 = 3,75\ kN/m^2$

$q = 3\ kN/m^2$ (**??**, tabela 10)

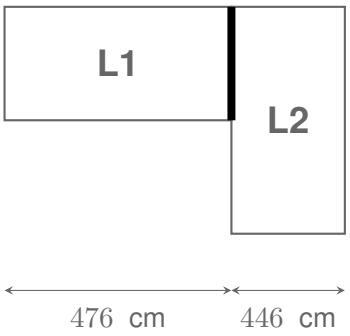
$g + q = 3,75 + 3 = 6,75\ kN/m^2$

3. Vinculação das lajes

   Multiplica por três por causa da recomendação do livro IBRACON.

   L1

   a) Vizinhança V5

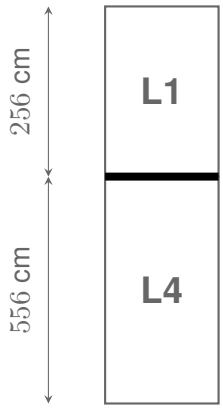


$\ell_1 < 3\ell_2$

$476 < 3 \times 446$

$476 < 1338 \rightarrow$ ambas engastadas

b) Vizinhança V2



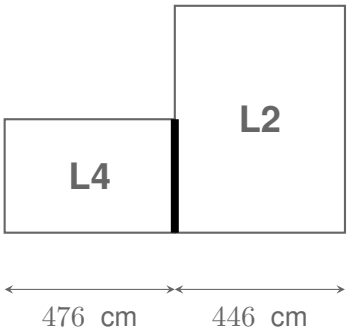
$\ell_1 < 3\ell_2$

$556 < 3 \times 256$

$556 < 768 \rightarrow$ ambas engastadas

L2

a) Vizinhança V5



$\ell_1 < 3\ell_2$

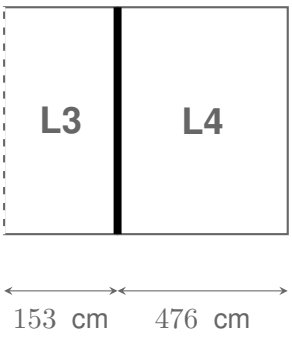
$476 < 3 \times 446$

$476 < 1138 \rightarrow$ ambas engastadas

L3

Por ser uma laje em balanço precisa de engaste.

a) Vizinhança V4



$\ell_1 < 3\ell_2$

$476 < 3 \times 153$

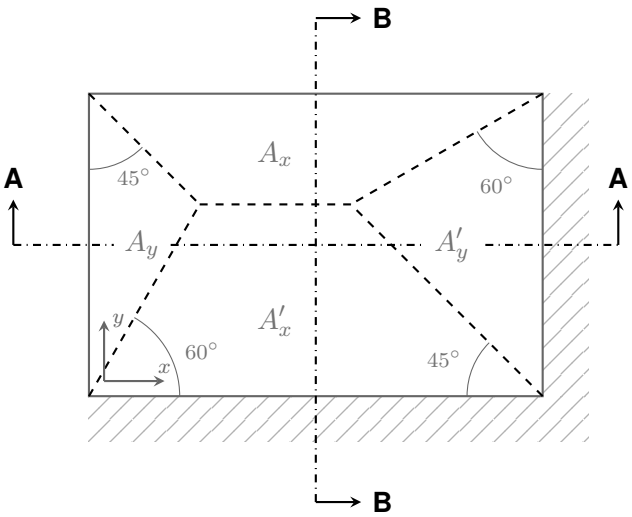
$476 < 459 \rightarrow$ ambas engastadas

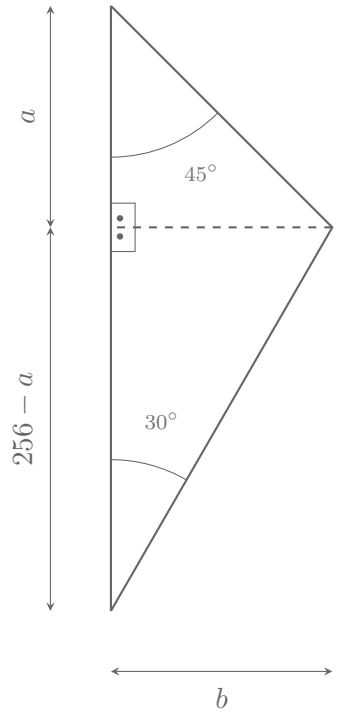
Maior laje fica apoiada e a menor fica engastada.

4. Reações verticais nas lajes

a) Charneiras plásticas

L1





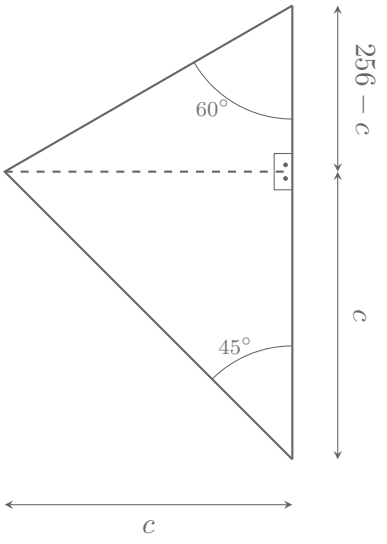
$$\cancel{\tan 45} = 1 \ \frac{b}{a} \longleftrightarrow b = a$$

$$\tan 30 = \frac{a}{256 - a}$$
$$\frac{\sqrt{3}}{3}(256 - a) = a$$
$$\frac{256\sqrt{3}}{3} = (1 + \frac{\sqrt{3}}{3})a$$
$$a \approx 93,70\ cm$$

$$L = 256 - a$$
$$L \approx 256 - 93,70$$
$$L \approx 162,30\ cm$$

$$A_y = \frac{bh}{2} \approx \frac{256 \times 93,7}{2} \approx 1,2\ m^2$$

$$V_y = \frac{(g+q)A_y}{\ell_x} \approx \frac{6,75 \times 1,2}{2,56} \approx 3,16\ kN/m$$

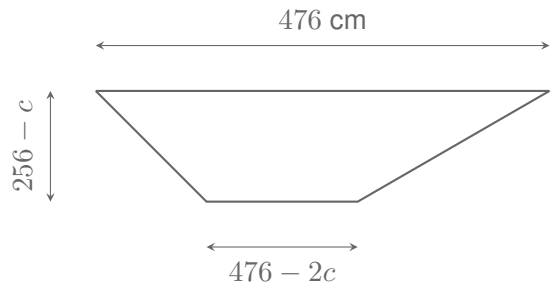


$$\tan 60 = \frac{c}{256 - c}$$
$$256\sqrt{3} = (1 + \sqrt{3})c$$
$$c \approx 162,30\ cm$$

$$L = 256 - c$$
$$\approx 256 - 162,30$$
$$\approx 93,70\ cm$$

$$A'_y = \frac{bh}{2} \approx \frac{256 \times 162,30}{2} \approx 2,08\ m^2$$

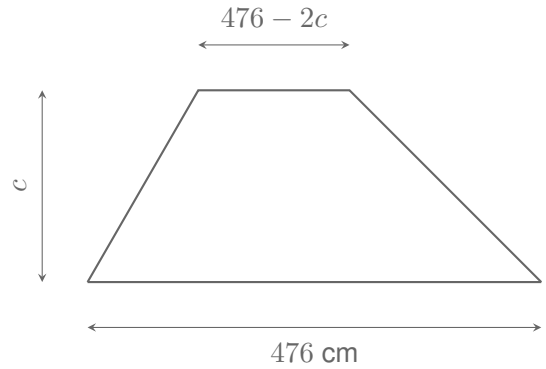
$$V'_y = \frac{(g+q)A'_y}{\ell_x} = \frac{6,75 \times 2,08}{2,56} \approx 5,48\ kN/m$$



$$b = 476 - 2c \approx 476 - 93,7 - 162,3 = 220\ cm$$

$$A_x = \frac{(B+b)h}{2} \approx \frac{(476+220)93,7}{2} \approx 3,26\ m^2$$

$$V_x = \frac{(g+q)A_x}{\ell_y} \approx \frac{6,75 \times 3,26}{4,76} \approx 4,62\ kN/m$$



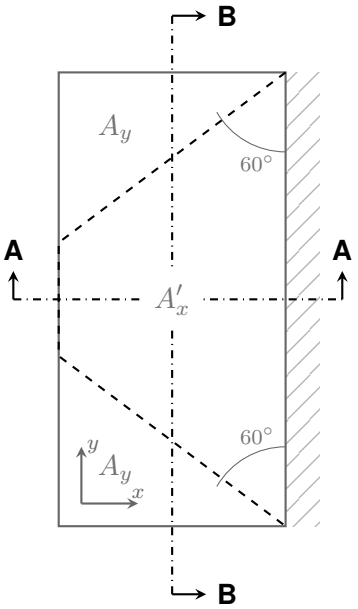
$$A'_x = \frac{(B+b)h}{2} \approx \frac{(476+220)162,3}{2} \approx 6,65\ m^2$$

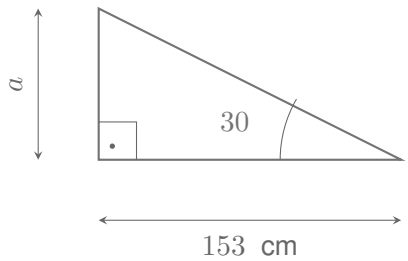
$$V'_x = \frac{(g+q)A_x}{\ell_y} \approx \frac{6,75 \times 5,65}{4,76} \approx 8,01\ kN/m$$

Continuação - para casa

L2

L3





$$\tan 30 = \frac{a}{1,53}$$
$$a \approx 0,88\ m$$

$$A_y = \frac{bh}{2} \approx \frac{1,53 \times 0,88}{2} \approx 0,68\ m^2$$

$$V_y = \frac{(g+q)A_y}{\ell_x} \approx \frac{6,75 \times 0,68}{1,53} \approx 2,98\ kN/m$$

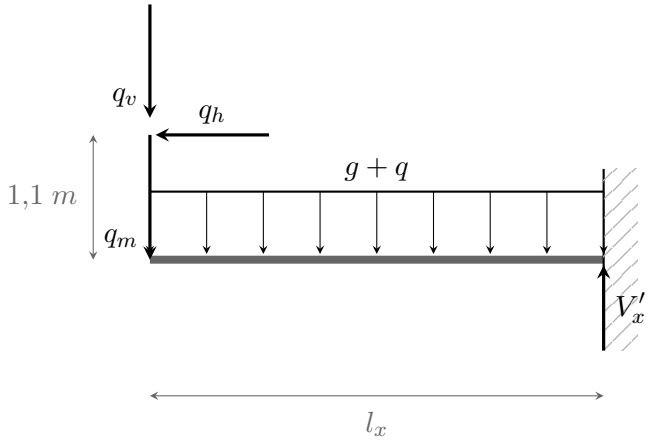


Figura 2 – Corte AA.

$g_m = 1\ kN/m$ Valor estimado da mureta (**??**, item 6.3)

$q_v = 2\ kN/m$ carga acidental (**??**, tabela 10, índice j)

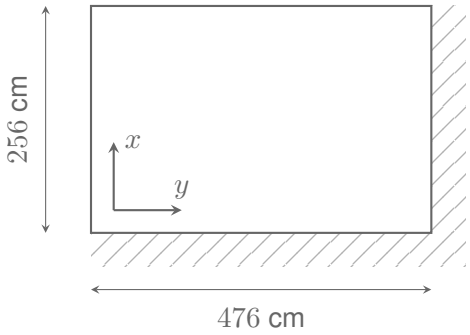
$q_h = 1\ kN/m$ (**??**, tabela 12)

$$V'_x = (g+q)\ell_x + g_m + q \rightarrow g_m = 1,1\ h_m$$
$$V'_x \approx 6,75 \times 1,53 + 1,1 \times 1 + 2$$
$$V'_x \approx 13,43\ kN/m$$

L4

b) Método tabelado

L1



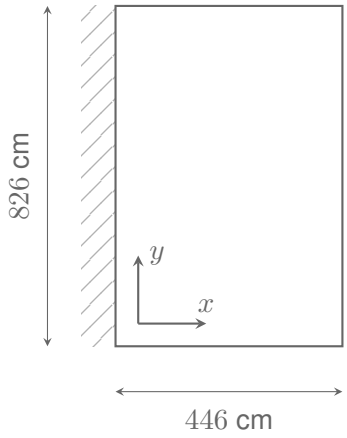
Laje tipo 3 (**??**, tabela 2.2b)

Por interpolação linear:

λ	v_x	v'_x	v_y	v'_y
1,86	3,17	4,64	2,17	2,17

$$V_x = v_x(g+q)\frac{l_x}{10} = 3,17(6,75)\frac{2,56}{10} = 5,47\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 8,01\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 3,75\ kN/m$$
$$V'_y = v'_y(g+q)\frac{l_x}{10} = 5,48\ kN/m$$

L2



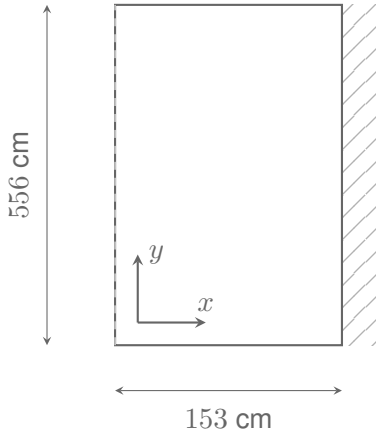
Laje tipo 2B (**??**, tabela 2.2a)

Por interpolação linear:

λ	v_x	v'_x	v_y
1,85	3,47	5,09	1,83

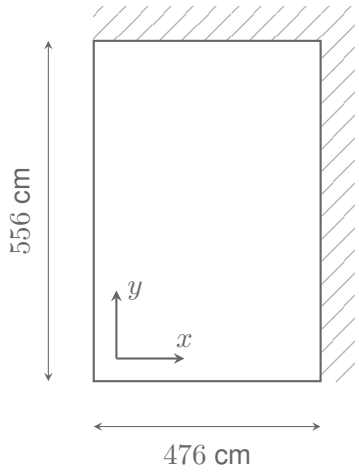
$$V_x = v_x(g+q)\frac{l_x}{10} = 10,45\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 15,33\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 5,51\ kN/m$$

L3



Não dá para fazer por tabela, tem que ser por charneiras plásticas.

L4



Laje tipo 3 (**??**, tabela 2.2b)

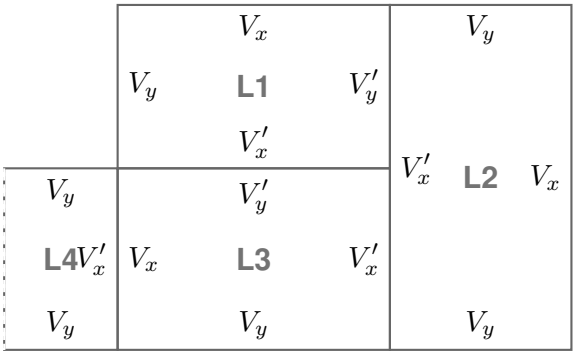
Por interpolação linear:

λ	v_x	v'_x	v_y	v'_y
1,17	2,48	3,62	2,17	3,17

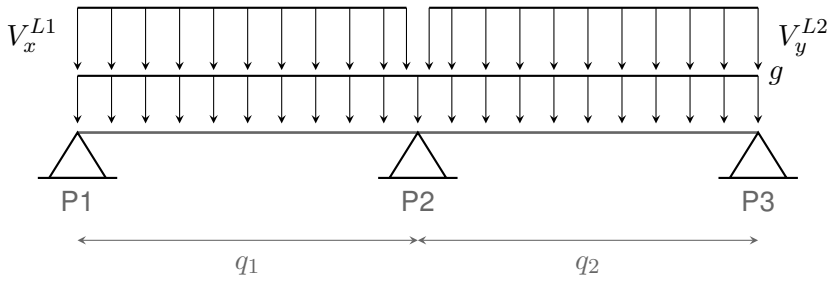
$$V_x = v_x(g+q)\frac{l_x}{10} = 7,96\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 11,64\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 6,97\ kN/m$$
$$V'_y = v'_y(g+q)\frac{l_x}{10} = 10,19\ kN/m$$

Aula 03a — 02 de setembro de 2026

5. Determinação do carregamento atuando nas vigas



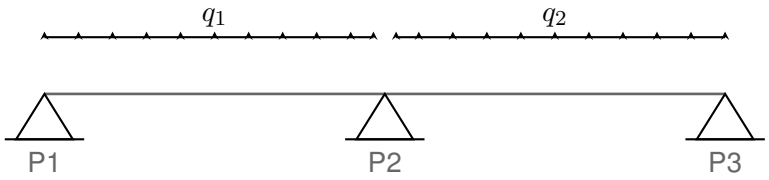
V1



$g = \gamma_{ap} \cdot A_{viga} \rightarrow \gamma_{ap} = 25\ kN/m^3$ (**??**, tabela 1)

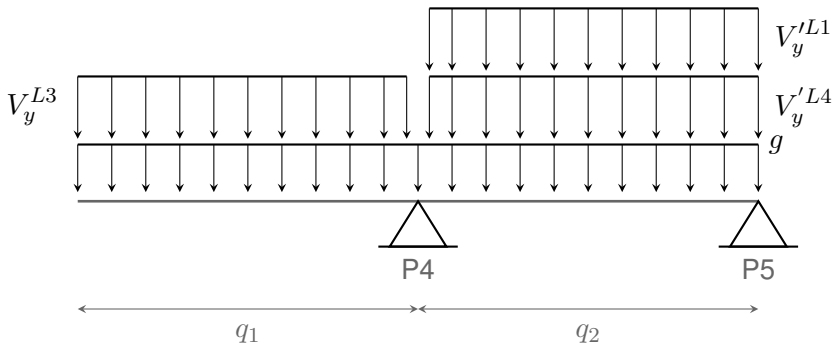
$q_1 = g + V_x^{'L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,47 = 7,72\ kN/m$

$q_2 = g + V_y^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 1,83 = 4,08\ kN/m$



Continuação - para casa

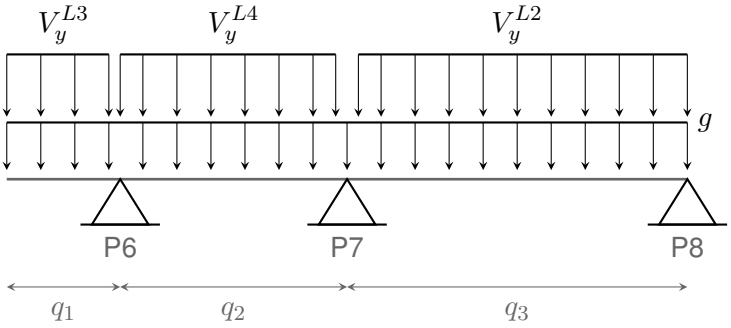
V2



$$q_1 = g + V_x^{\prime L3} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3 = 5,25\ kN/m$$

$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 10,19 + 5,48 = 17,92\ kN/m$$

V3

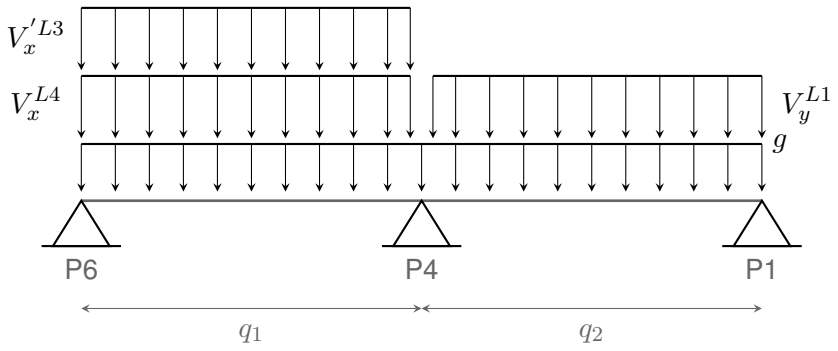


$$q_1 = g + V_x^{\prime L3} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3 = 5,25\ kN/m$$

$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 6,97 = 9,22\ kN/m$$

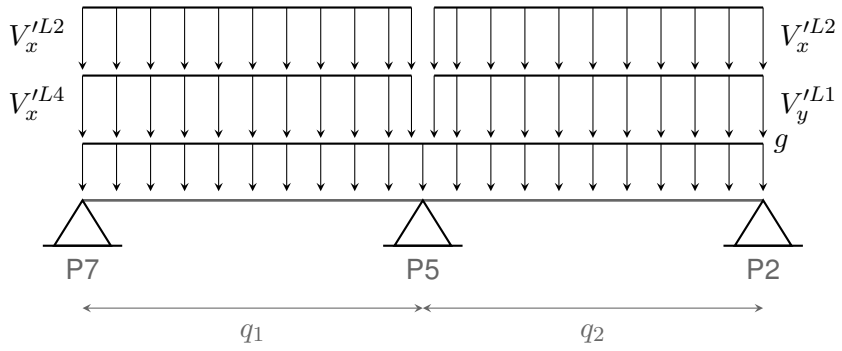
$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,51 = 7,76\ kN/m$$

V4



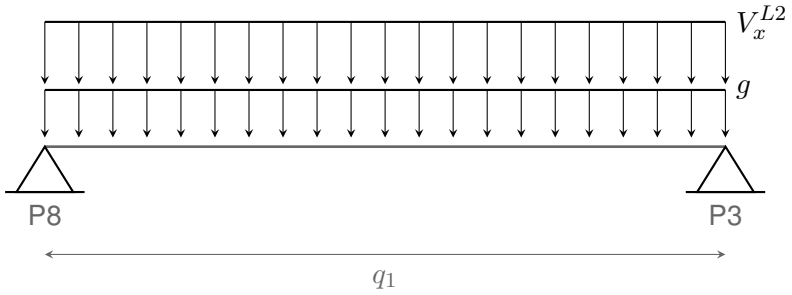
$$q_1 = g + V_x^{'L4} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 11,78 + 7,96 = 21,99\ kN/m$$
$$q_2 = g + V_y^{'L1} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3,75 = 6\ kN/m$$

V5



$$q_1 = g + V_x^{'L4} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 11,64 + 15,33 = 29,22\ kN/m$$
$$q_2 = g + V_y^{'L1} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,48 + 15,33 = 23,06\ kN/m$$

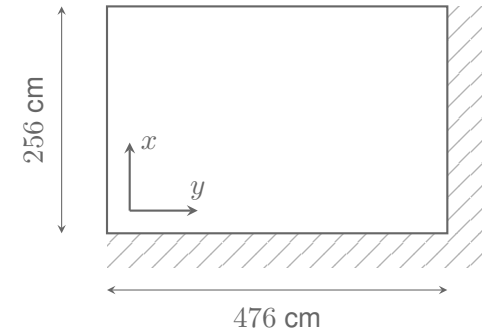
V6

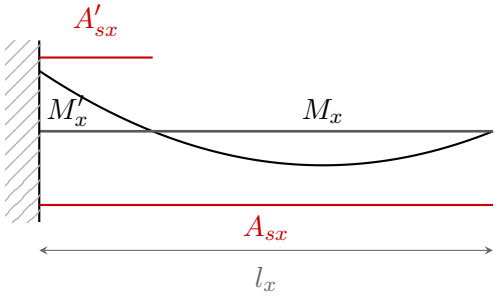


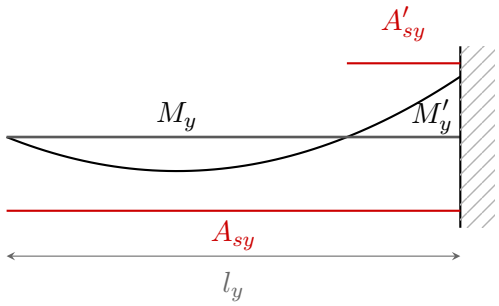
$$q_1 = g + V_x^{L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 10,45 = 14,2\ kN/m$$

6. Esforços solicitantes - momentos fletores

L1







Laje tipo 3 (**??**, tabela 2.3b)

λ	μ_x	μ'_x	μ_y	μ'_y
1,86	5,50	11,57	1,71	8,17

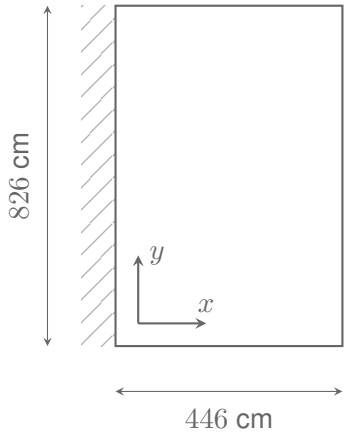
$$M_x = \mu_x(g+q)\frac{\ell_x^2}{100} = 5,50 \times 6,75\ \frac{2,56^2}{100} = 2,43\ kNm/m$$

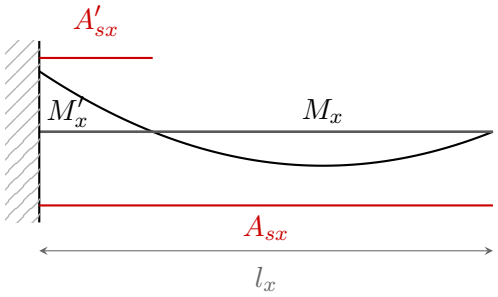
$$M'_x = \mu'_x(g+q)\frac{\ell_x^2}{100} = 5,12\ kNm/m$$

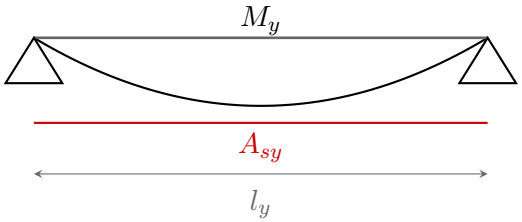
$$M_y = \mu_y(g+q)\frac{\ell_x^2}{100} = 0,76\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x^2}{100} = 3,61\ kNm/m$$

L2







Laje tipo 2B (**??**, tabela 2.3a)

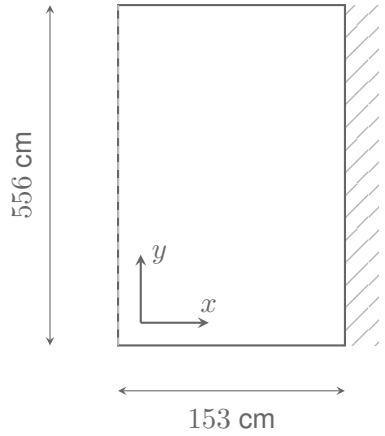
λ	μ_x	μ'_y	μ'_y
1,85	5,81	11,94	1,67

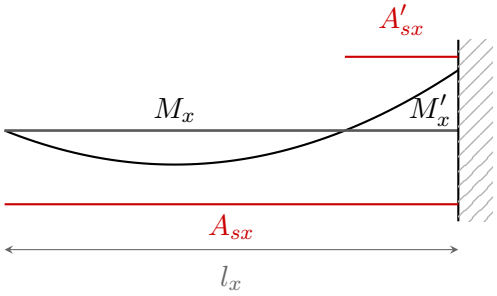
$$M_x = \mu_x(g+q)\frac{\ell_x^2}{100} = 7,80\ kNm/m$$

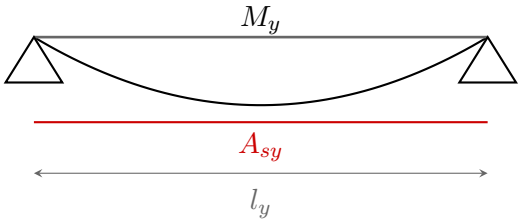
$$M'_x = \mu'_x(g+q)\frac{\ell_x^2}{100} = 16,04\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x^2}{100} = 2,24\ kNm/m$$

L3







Laje tipo 9 (**??**, tabela 2.3e)

$$\gamma = \frac{\ell_a}{\ell_b} = \frac{1,53}{5,56} = 0,28$$

γ	μ_x	μ'_x	μ_y	μ_{yb}
0,28	-12,50	50	0,78	6,22

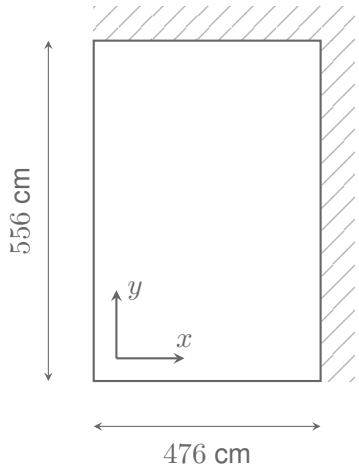
$$M_x = \mu_x \; g \; \frac{\ell_x^2}{100} = -12,50 \times 3,75 \; \frac{1,53^2}{100} = -1,10 \; kNm/m$$

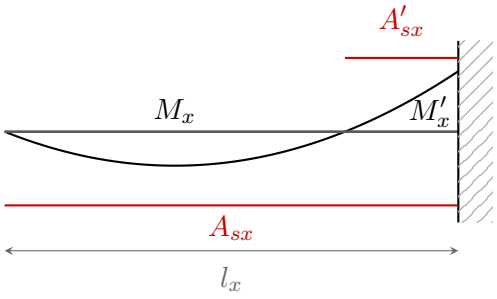
$$M'_x = \mu'_x(g+q)\frac{\ell_a^2}{100} + (g_m + q_v)\;\ell_a + q_h \times L_m = 13,59 \; kNm/m$$

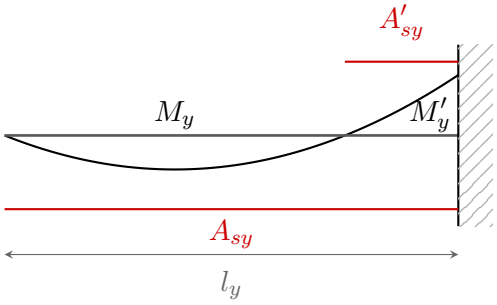
$$M_y = \mu_y \; g\frac{\ell_a^2}{100} = 0,78 \times 3,75 \times \frac{1,53^2}{100} = 0,07 \; kNm/m$$

$$M_{yb} = \mu_{yb}(g+q)\frac{\ell_x^2}{100} = 6,22 \times 6,75 \; \frac{1,53^2}{100} = 0,98 \; kNm/m$$

L4







Laje tipo 3 (**??**, tabela 2.3b)

λ	μ_x	μ'_x	μ_y	μ'_x
1,17	3,51	8,43	2,44	7,55

$$M_x = \mu_x(g+q)\frac{\ell_x}{100} = 5,36\ kNm/m$$

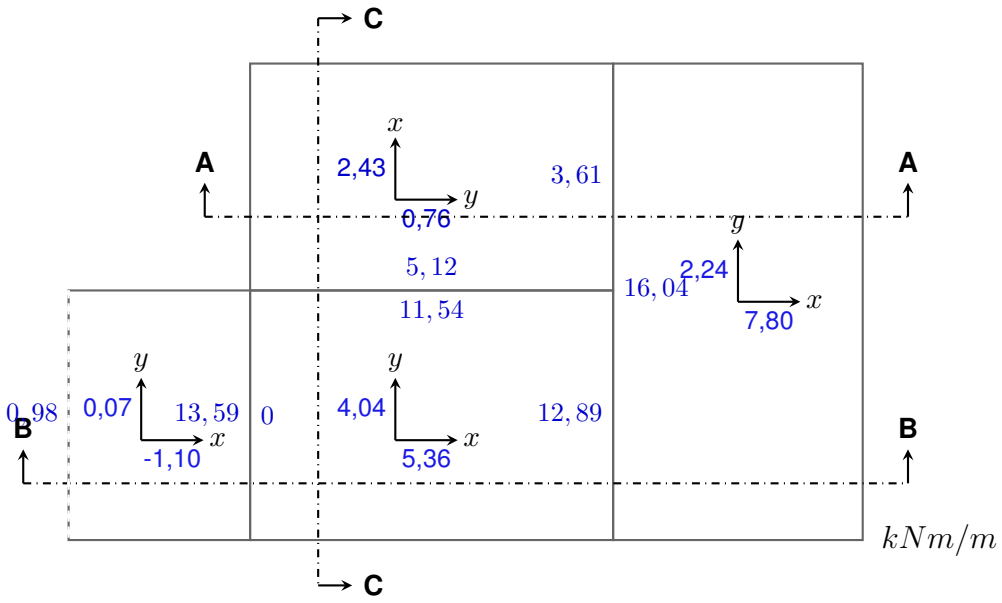
$$M'_x = \mu'_x(g+q)\frac{\ell_x}{100} = 12,89\ kNm/m$$

$$M_y = \mu_y(g+q)\frac{\ell_x}{100} = 4,04\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x}{100} = 11,54\ kNm/m$$

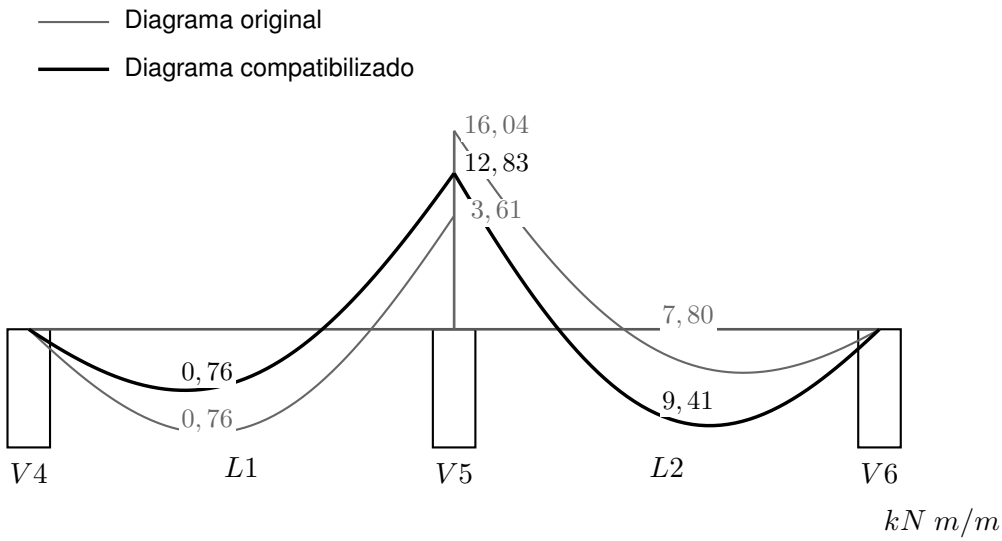
Aula 03b — 03 de setembro de 2026

7. Compatibilização dos momentos fletores



a) Método por cálculo

i. Corte AA

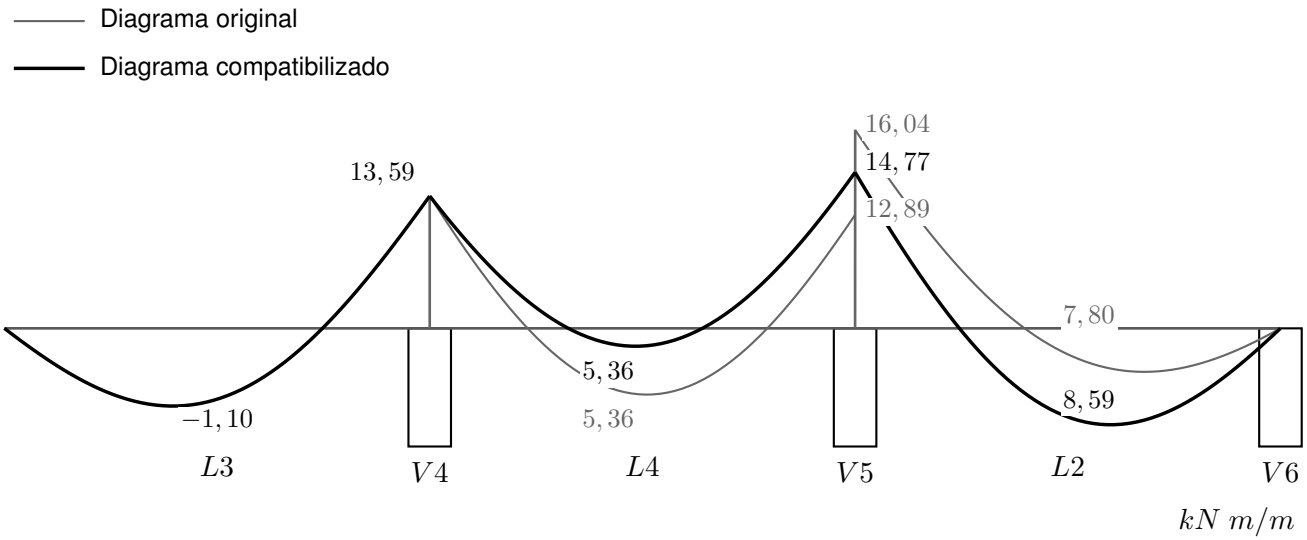


Não precisa fazer a compatibilização da L1 para ficar a favor da segurança.
refendobservacao

$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{16,04 + 3,61}{2} = 9,83\ kNm/m \\ 0,8M' = 0,8 \times 16,04 = 12,83\ kNm/m \end{cases}$$

$$M_{c,\ L2} = M_{eixo} + \frac{M' - M_{med}}{2} = 7,8 + \frac{16,04 - 12,83}{2} = 9,41\ kNm/m$$

ii. Corte BB

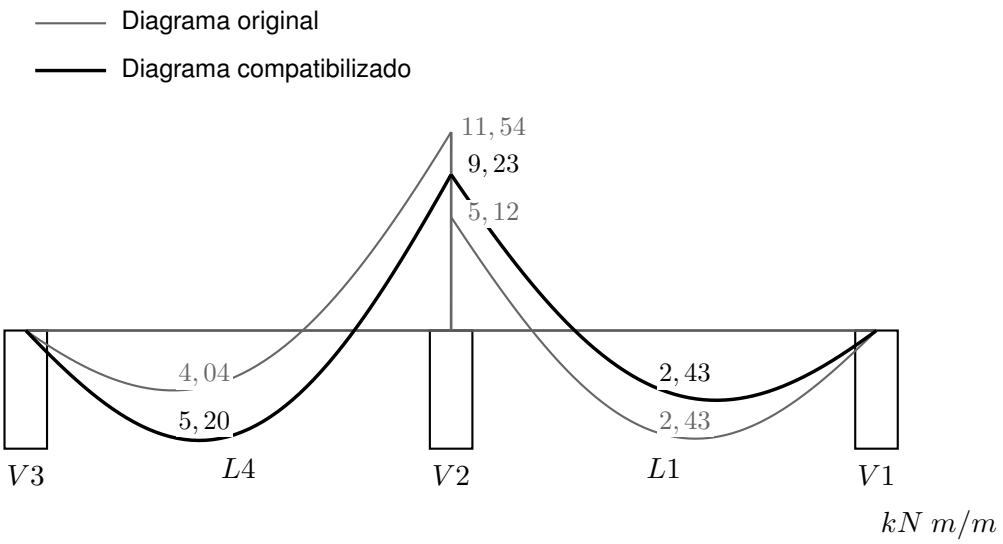


Como a L3 é em balanço, os valores se mantém.

$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{16,04 + 12,89}{2} = 14,77\ kNm/m \\ 0,8M' = 0,8 \times 16,04 = 12,83\ kNm/m \end{cases}$$

$$M_{c,\ L2} = M_{eixo} + \frac{M' - M_{med}}{2} = 7,8 + \frac{16,04 - 14,77}{2} = 8,59\ kNm/m$$

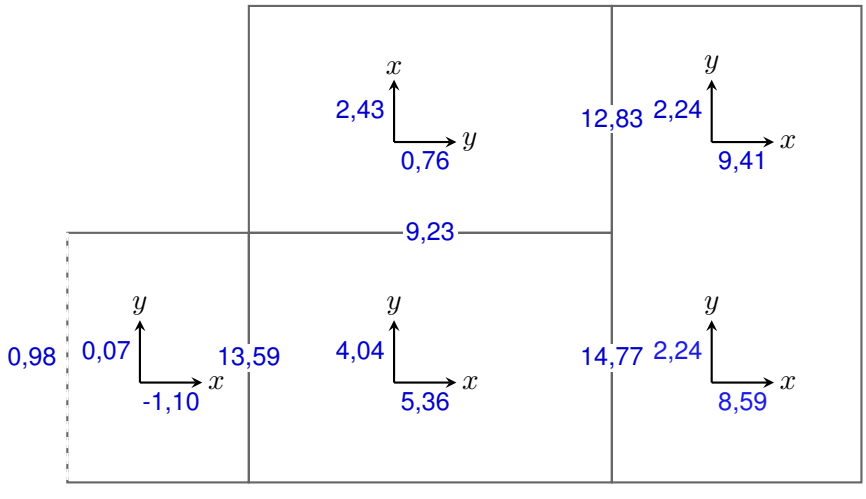
iii. Corte CC



Não precisa fazer a compatibilização da L1 para ficar a favor da segurança.

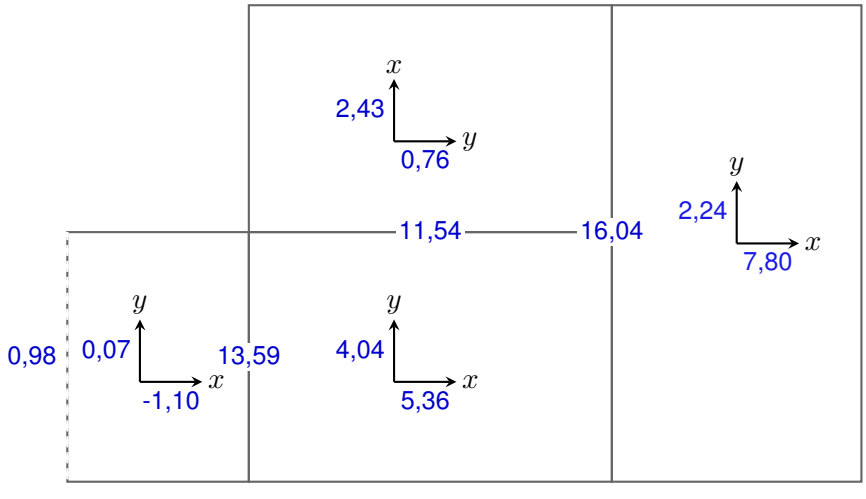
$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{11,54 + 5,12}{2} = 8,83\ kNm/m \\ 0,8M' = 0,8 \times 11,54 = 9,23\ kNm/m \end{cases}$$

$$M_{c,\ L4} = M_{eixo} + \frac{M' - M_{med}}{2} = 4,04 + \frac{11,54 - 9,23}{2} = 5,20\ kNm/m$$



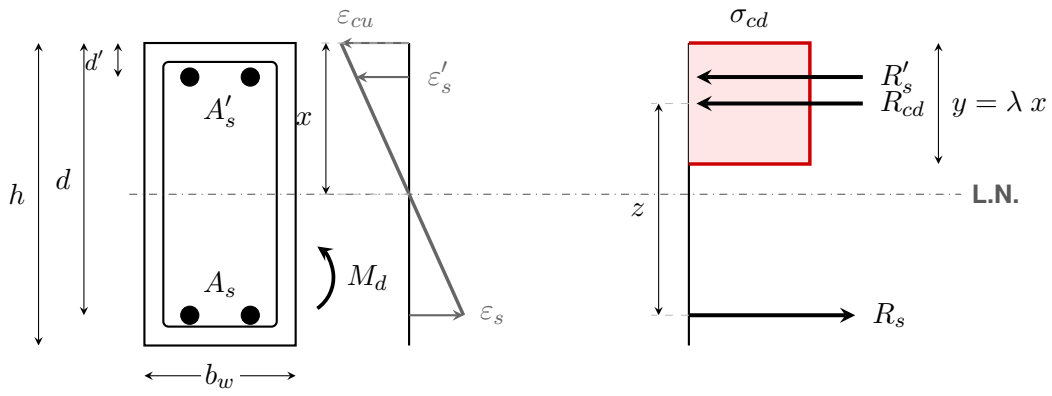
kNm/m

b) Método ABNT



kNm/m

Dimensionamento a flexão por tabelas Kc e Ks



Equilíbrio de momento

$$M_d = R_c\left(d - \frac{y}{2}\right) + R's(d - d'))$$

$$M_d = (b_w\ y\ \sigma_{cd})\left(d - \frac{y}{2}\right) + (A's\ \sigma'_s)(d - d')$$

$$M_d = (b_w\ \lambda\ x\ \alpha_c\ \eta_c\ f_{cd})\left(d - \frac{\lambda\ x}{2}\right) + (A's\ \sigma'_s)(d - d') \rightarrow \beta_x = \frac{x}{d} \leftrightarrow x = \beta_x\ d$$

$$M_d = b_w\ \lambda\ \beta_x\ d\ \alpha_c\ \eta_c\ f_{cd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right) + \cancelto{0}{(A's\ \sigma'_s)(d - d')}$$

$$M_d = b_w\ \lambda\ \beta_x\ d^2\ \alpha_c\ \eta_c\ f_{cd}\left(1 - \frac{\lambda\beta_x}{2}\right)$$

$$\rightarrow f_{ck} = 40\ MPa\ (\lambda = 0,8, \eta_c = 1, \alpha_c = 0,85)$$

$$M_d = b_w\ \lambda\ \beta_x\ d^2\ \alpha_c\ \eta_c\ f_{cd}\left(1 - \frac{\lambda\beta_x}{2}\right)$$

$$M_d = 0,68\ b_w\ d^2\ f_{cd}\ \beta_x(1 - 0,4\beta_x) \longleftrightarrow \frac{b_w\ \cancel{d^2}}{M_d} = \frac{1}{0,68\ \beta_x\ f_{cd}(1 - 0,4\beta_x)} \quad K_c = f(\beta_x, f_{cd})$$

Do equilíbrio de tensões

$$M_d = R_s\left(d - \frac{y}{2}\right)$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ x}{2}\right) \rightarrow \beta_x = \frac{x}{d} \leftrightarrow x = \beta_x\ d$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right)$$

$$\rightarrow f_{ck} < 50\ MPa(\lambda = 0,80)$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right)$$

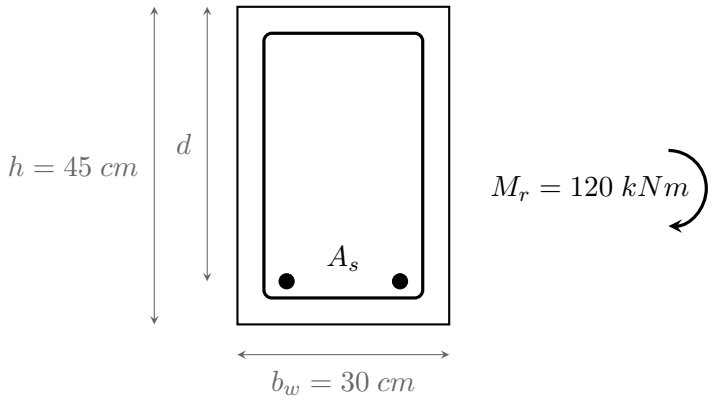
$$M_d = A_s\ \sigma_{sd}\ d\left(1 - 0,4\ \beta_x\right)$$

$$K_s = f(\sigma_s, \beta_x)$$

$$\frac{A_s\ \sigma_{sd}}{M_d} = \frac{1}{1 - 0,4\ \beta_x}$$

Exemplo

1. Calcule a área de aço. Dados: C25, aço CA-50, CAA-II.



a) Momento de projeto

$\gamma_c = 1,4$ (**??**, tabela 12.1)

$$M_d = \gamma_c\, M_r = 1,4 \times 120 = 168\ kNm = 16\,800\ kN\ cm$$

b) Altura útil

$$d = h - c - \phi_e - n\frac{\phi_\ell}{2} = 45 \cdot 10^{-2} - 25 \cdot 10^{-3} - 5 \cdot 10^{-3} - 2\ \frac{10 \cdot 10^{-3}}{2} = 41\ cm$$

c) Kc

$$K_c = \frac{b_w\, d^2}{M_d} = \frac{30 \times 41^2}{16800} = 3\ cm^2/kN$$

d) Tipo de armadura

$\beta_x = 0,32 \leq 0,45 \rightarrow$ armadura simples, domínio 3 (**??**, tabela 1.1a)

e) Ks

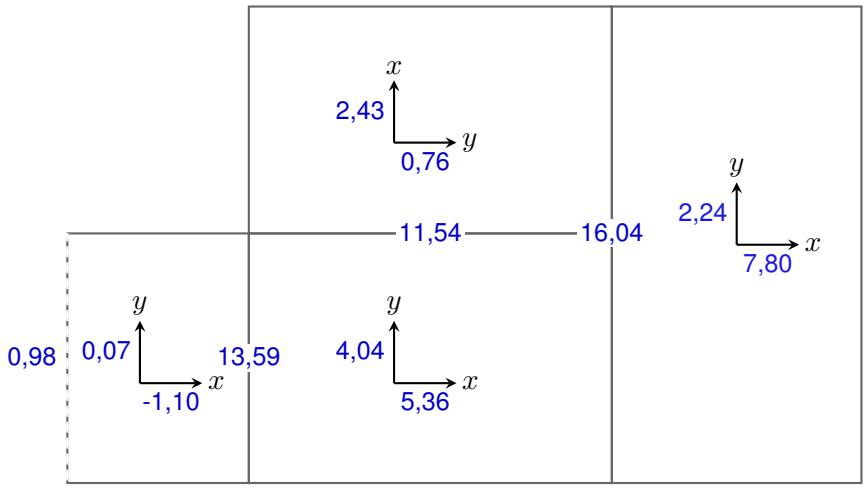
$K_s = 0,026$ (**??**, tabela 1.1a)

f) Área de aço

$$K_s = \frac{A_s\, d}{M_d} \longleftrightarrow A_s = \frac{K_s\, M_d}{d} = \frac{0,026 \times 16800}{41} = 10,65\ cm^2$$

Aula 04a — 09 de setembro de 2026

8. Dimensionamento armadura positiva



kNm/m

L1

a) Momento de projeto

$\gamma_c = 1,4$ (**??**, tabela 12.1)

$$M_{d,x} = \gamma_c M_x = 1,4 \times 2,43 = 340,2\ kNcm/m$$
$$M_{d,y} = \gamma_c M_y = 1,4 \times 0,76 = 106,4\ kNcm/m$$

b) Altura útil

CAA-I = 20 mm (laje) (**??**, tabela 7.2)

$$d = h - c - \phi_e - \frac{\phi_\ell}{2} = 10 \cdot 10^{-2} + \left(-20 - 10 - \frac{10}{2} \right) \cdot 10^{-3} = 6,5\ cm$$

c) Taxa mínima de armadura

$\rho_s = \dfrac{A_s}{A_c} \rightarrow A_c = bh$ (**??**, tabela 19.1)

$\rho_{min} = 0,179\%$ (**??**, tabela 17.3)

$\rho_s \geq 0,67\rho_{min}$(**??**, tabela 19.1)

$$A_{s,min} \geq 0,67\ \frac{0,179 \times 100 \times 10}{100}$$
$$A_{s,min} \geq 1,2\ cm^2$$

d) Área de aço

$$K_{c,x} = \frac{b\ d^2}{M_d} = \frac{100 \times 6,5^2}{340,2} = 12,42\ cm^2/kN$$

$$K_{c,y} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{106,4} = 39,71\ cm^2/kN$$

$\beta_{x,x} = 0,06 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)
$\beta_{x,y} = 0,02 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)

$$\begin{cases} A_s \geq A_{s,min} \rightarrow A_s \\ A_s < A_{s,min} \rightarrow A_{s,min} \end{cases}$$

$$K_{s,x} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,024 \times 340,2}{6,5} = 1,26\ cm^2/m$$
$$K_{s,y} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,023 \times 106,4}{6,5} = 0,38\ cm^2/m$$

e) Determinação da bitola e do espaçamento

$\phi_\ell \leq \frac{h}{8} = \frac{10}{8} = 1,25\ cm = 12,5\ mm$ (**??**, item 20.1)

(**??**, item 20.1)

$$S_x \leq \begin{cases} 2h = 2 \times 10 = 20\ cm \\ 20\ cm \end{cases} \qquad S_y \leq 33\ cm$$

$$A_{\phi,x} = \pi\left(\frac{D}{2}\right)^2 = \pi\left(\frac{5 \cdot 10^{-3}}{2}\right)^2 = 0,196\ cm^2$$

O espaçamento deve ser arredondado para baixo.

$$S_{\phi,x} = \frac{A_\phi}{A_s}\,100 = \frac{0,196}{1,26}\,100 = 15,58 \approx 15\ cm$$
$\phi_{\ell,x} = 5\ mm$ c/ $15\ cm$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\,100 = \frac{0,196}{1,2}\,100 = 16,36 \approx 16\ cm$$
$\phi_{\ell,y} = 5\ mm$ c/ $16\ cm$

L2

a) Momento de projeto

$M_{d,x} = \gamma_c M_x = 1,4 \times 7,80 = 1092\ kNcm/m$
$M_{d,y} = \gamma_c M_y = 1,4 \times 2,24 = 313,6\ kNcm/m$

b) Área de aço

$$K_{c,x} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{1092} = 3,87\ cm^2/kN$$

$$K_{c,y} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{313,6} = 13,47\ cm^2/kN$$

$\beta_{x,x} = 0,14 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)
$\beta_{x,y} = 0,02 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)

$$K_{s,x} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,025 \times 1092}{6,5} = 4,20\ cm^2/m$$
$$K_{s,y} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,023 \times 313,6}{6,5} = 1,11\ cm^2/m$$

c) Determinação da bitola e do espaçamento

$$A_{\phi,x} = \pi\left(\frac{D}{2}\right)^2 = \pi\left(\frac{8 \cdot 10^{-3}}{2}\right)^2 = 0,5\ cm^2$$
$$S_{\phi,x} = \frac{A_\phi}{A_s}\,100 = \frac{0,5}{4,2}\,100 = 11,98 \approx 11\ cm$$
$\phi_{\ell,x} = 8\ mm$ c/ $11\ cm$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\,100 = \frac{0,196}{1,2}\,100 = 16,36 \approx 16\ cm$$
$\phi_{\ell,y} = 5\ mm$ c/ $16\ cm$

L3

Corrigir

L4

a) Momento de projeto

$$M_{d,x} = \gamma_c M_x = 1,4 \times 5,36 = 750,4\ kNcm/m$$
$$M_{d,y} = \gamma_c M_y = 1,4 \times 4,04 = 565,6\ kNcm/m$$

b) Área de aço

$$K_{c,x} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{750,4} = 5,63\ cm^2/kN$$
$$K_{c,y} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{565,6} = 7,47\ cm^2/kN$$

$\beta_{x,x} = 0,10 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)
$\beta_{x,y} = 0,08 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (**??**, tabela 1.1a)

$$K_{s,x} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,024 \times 750,4}{6,5} = 2,77\ cm^2/m$$

$$K_{s,y} = \frac{A_s\, d}{M_d} \longleftrightarrow A_s = \frac{K_s\, Md}{d} = \frac{0,024 \times 565,6}{6,5} = 2,09\ cm^2/m$$

c) Determinação da bitola e do espaçamento

$$A_{\phi,x} = \pi\left(\frac{D}{2}\right)^2 = \pi\left(\frac{8 \cdot 10^{-3}}{2}\right)^2 = 0,5\ cm^2$$

$$S_{\phi,x} = \frac{A_\phi}{A_s}\ 100 = \frac{0,5}{2,77}\ 100 = 18,05 \approx 18\ cm$$

$$\phi_{\ell,x} = 8\ mm \text{ c/ } 18,05\ cm$$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\ 100 = \frac{0,5}{2,09}\ 100 = 23,92 \approx 23\ cm$$

$$\phi_{\ell,y} = 8\ mm \text{ c/ } 23\ cm$$